



A submission for the NASA Space Apps Challenge 2025.

CosmicZoom: Exploring the Universe, pixel by pixel

Nasa SpaceApps Challenge:

Embiggen Your Eyes!

Overwhelming Datasets

NASA generates massive image files—often in the gigapixel to terapixel range—that are impractical for standard devices or home internet connections.

Accessibility Barrier

The sheer size of these datasets prevents public users and even many researchers from easily exploring the detailed visual information within them.

Inefficient Exploration

Traditional methods require downloading entire files or rely on static, pre-rendered views, losing the immersive experience of discovery.

Introducing CosmicZoom: Adaptive Exploration

CosmicZoom is a web application designed to turn overwhelming astronomical datasets into accessible, interactive maps of space.



Interactive Gigapixel Views

Pan and zoom across massive images (up to terapixel scale) directly in your browser, without heavy downloads. It offers



Adaptive Loading Engine

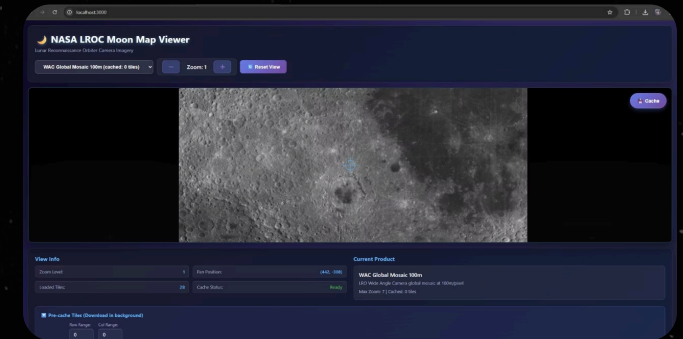
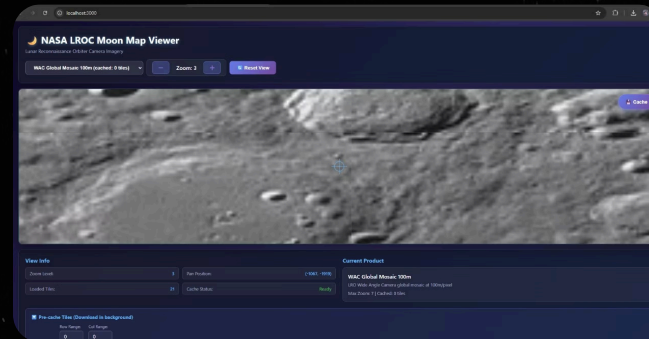
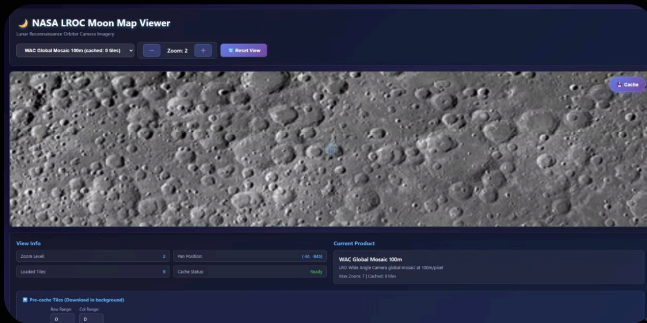
Only the visible image tiles are loaded, ensuring smooth performance even on standard devices like mobile phones or tablets.



Precision Searching

Search astronomical images by precise coordinates, turning a vast image into a searchable resource.

- The user interface allows zooming upto 7 levels and seamless panning.
- It allows the user to compare mutiple datasets as well.



How CosmicZoom Works: The Technical Core

Our solution leverages a tile-based serving architecture (like WebGL or OpenSeadragon) combined with standard web technologies for maximum compatibility.

Data Storage

Gigapixel source images are stored on backend servers and pre-processed into a multi-resolution pyramid of smaller, optimized image tiles.

Efficient Delivery

The backend server serves only those specific, low-latency image tiles, enabling rapid visualization.



User Interaction

The user pans or zooms via the web interface (HTML/JavaScript).

Tile Request

The front end calculates which tiles at which resolution are needed for the current viewport.

Impact & Future Vision

CosmicZoom's innovative approach has dual applications, significantly lowering the barrier to entry for space data.

Public Engagement & Education

Imagine interactive displays in science museums or educational settings where visitors can explore the deepest corners of the universe on a simple touchscreen.

We make complex data visually thrilling.

Advanced Research Potential

With improvements like spectral data layer integration, CosmicZoom can become a vital tool for astronomers, allowing them to quickly survey large areas for targets before initiating full data downloads.



CosmicZoom: Exploring the Universe, pixel by pixel

Our vision is to transform NASA's overwhelming datasets into accessible, explorable maps of space, accelerating both public fascination and scientific insight.

Team

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